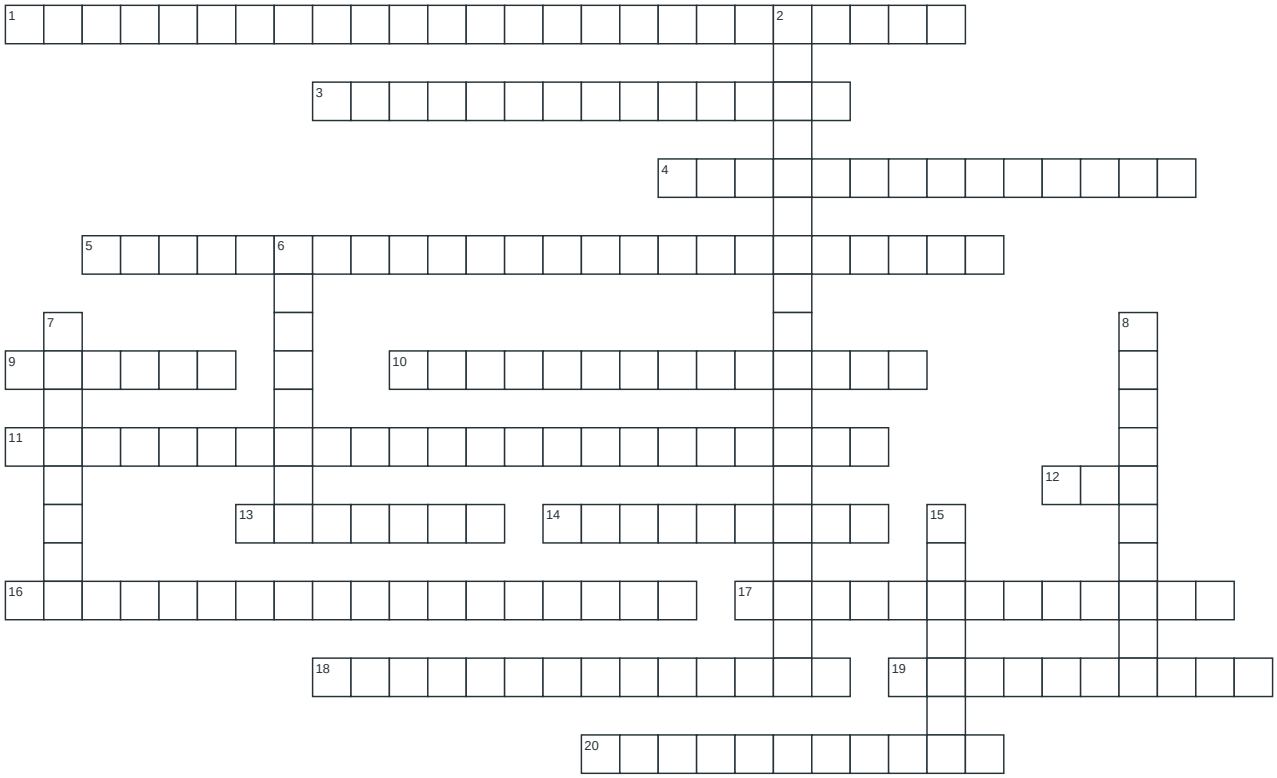
Biotechnology Crossword

Hard | 20 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

1. Membrane network with attached ribosomes that helps process proteins.

3. Plant-cell compartment that stores water and helps maintain pressure.

4. Boundary that regulates exchange between a cell and its surroundings.

5. Relationship between a cell's exchange surface and its internal volume.

9. Group of similar cells and associated material performing related functions.

10. Organelle that modifies, sorts, and packages many proteins.

11. Making many copies of a selected DNA region

12. Molecule that stores hereditary information in most cells

13. Small DNA molecule separate from a bacterium's main chromosome

14. Cell contents outside the nucleus and inside the plasma membrane.

ACROSS (continued)

16. Method that separates DNA fragments by size using an electric field

17. Use of organisms or biological processes to make useful products

18. DNA molecule assembled from genetic material from different sources

19. States that living things are made of cells and that cells come from existing cells.

20. Organelle where photosynthesis occurs in plants and algae.

DOWN

2. Process by which cells develop structures suited to particular functions.

6. Rigid outer layer that supports many plant, fungal, and bacterial cells.

7. Cell structure that assembles proteins.

8. Cell type that lacks a membrane-bound nucleus.

15. Membrane-bound organelle that contains most eukaryotic DNA.

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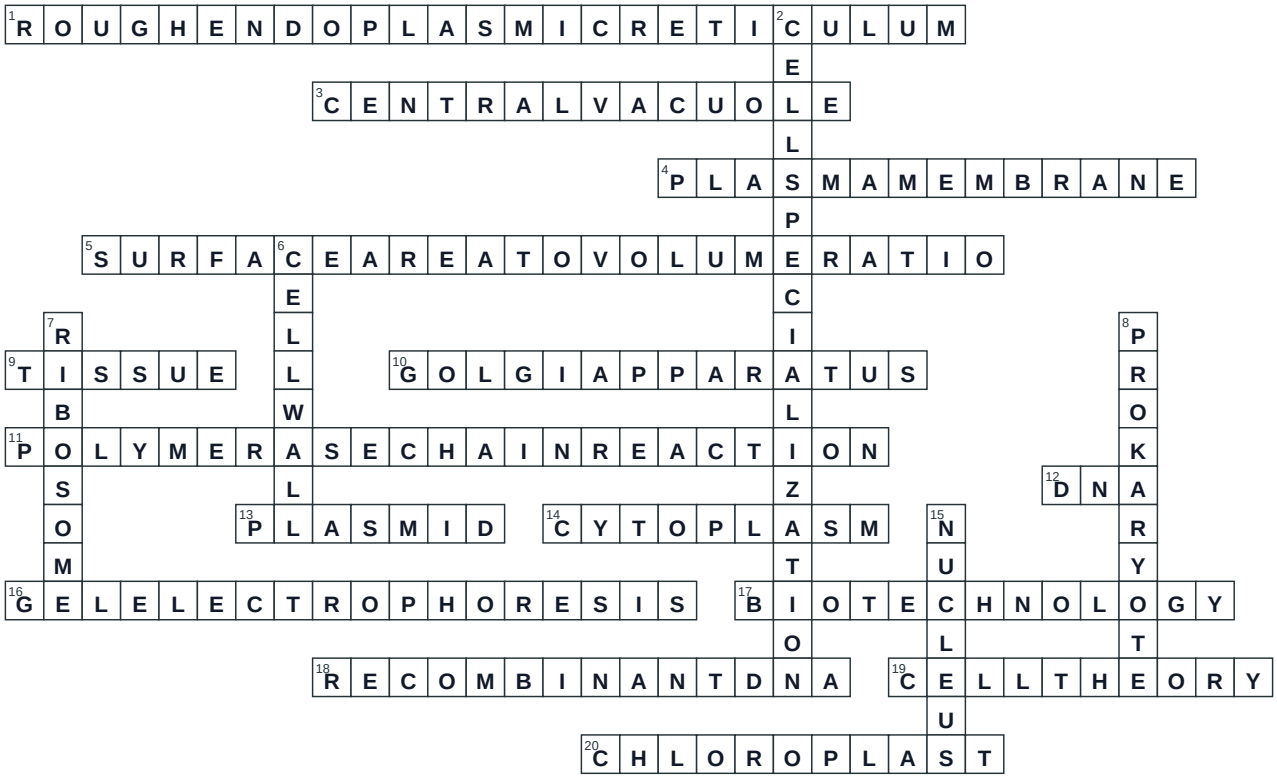
Hard | 20 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

1. Rough endoplasmic reticulum
3. Central vacuole
4. Plasma membrane
5. Surface area-to-volume ratio
9. Tissue
10. Golgi apparatus
11. Polymerase Chain Reaction
12. DNA
13. Plasmid
14. Cytoplasm
16. Gel Electrophoresis

ACROSS (continued)

17. Biotechnology
18. Recombinant DNA
19. Cell theory
20. Chloroplast

DOWN

2. Cell specialization
6. Cell wall
7. Ribosome
8. Prokaryote
15. Nucleus

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